Computer Networks and Internet Architecture

MCA104 Exam Preparation – Overview, Past-Paper Analysis and Study Plan

Exam: Wednesday 18 November 2026

Exam Focus

How to use this set. Five module PDFs plus this overview. Every past-paper question we have is placed in the module it belongs to, so as you revise a module you can see exactly how it has been asked before. Section 6 is a five-day plan; Section 7 is the formula sheet.

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The six papers we have

Paper	Marks	Format
Sept 2026 CAE-I	60	2 hours. Sections A, B, C, 10 marks each.
Oct 2026 CAE-II	30	1 hour. Section A, B, C, 10 marks each.
Oct 2025 CAE-II	20	60 minutes. Section A and B, 10 marks each.
Oct 2024 CAE-II	20	60 minutes. Section A and B, 10 marks each.
Oct 2025 full	75	3 hours. Section A: 5 questions $\times$ 15 marks, all compulsory.
Sept 2024 CAE-I	40	90 minutes. Section A and B, 10 marks each.

Exam Focus

The two most predictive papers are the Sept 2026 CAE-I and the Oct 2026 CAE-II, because they are from your own batch and their coverage matches the current syllabus. Read Section 3 for those first.

Marks by module

Module	Content	Papers hit
1	Data communication, OSI and TCP/IP, media, switching, multiplexing, error control, MAC	6 of 6
2	Network layer, addressing, subnetting, routing, IPv4, IPv6, ICMP	6 of 6
3	Transport layer, TCP, UDP, flow and congestion control, QoS	6 of 6
4	Application layer, DNS, DHCP, security, TLS, IPSec	5 of 6
5	Internet structure, ISP, IXP, AS, CDN, cloud	5 of 6

Module 1 carries the most marks overall because both CAE-I papers are almost entirely Module 1. **Module 3 and Module 4 carry the most marks in the semester paper.**

The recurring questions, module by module

Module 1

Paper	Question
Sept 2026	Explain the importance of the layered approach.
Sept 2026	Define data communication; five components; performance characteristics.
Sept 2026	Random access protocols: ALOHA, slotted ALOHA, CSMA, CSMA/CD.
Sept 2026	Transmission media: guided and unguided.
Sept 2026	Switching techniques; compare circuit and packet.
Sept 2026	TCP/IP suite versus OSI.
Sept 2026	Multiplexing: FDM, TDM, WDM.
Sept 2026	CRC for data 1101011011, generator 10011.
Oct 2025	Define multiplexing; FDM and TDM with diagrams.
Oct 2025	Data communication components; framing.
Oct 2025	Go-Back-N versus Selective Repeat.
Oct 2025	Flow control in the data link layer.
Oct 2025	TDM versus FDM.
Oct 2025 CAE-II	CRC for 10011101 with $x^3 + 1$; CSMA/CD scenario with 5 computers.
Oct 2024 CAE-II	CRC for 10011101; sliding window protocols; LLC versus MAC.
Sept 2024	Polar coding schemes; PCM; wireless versus wired; PSTN.

Module 2

Paper	Question
Sept 2026	RIP versus OSPF: algorithm, metric, convergence, scalability.
Sept 2026	Structure of an IPv4 datagram with a diagram.
Oct 2025	ICMP in troubleshooting: echo and destination unreachable.
Oct 2025	Responsibilities of the network layer; IPv6 versus IPv4.
Oct 2025	Functions of the network layer and its services to transport.
Oct 2025	Routing tables and dynamic routing; subnetting.
Oct 2024 CAE-II	ARP for MAC mapping failure.

Module 3

Paper	Question
Sept 2026	Connection-oriented versus connectionless; TCP and UDP as examples.
Sept 2026	Short notes: transport services, SSL/TLS, TCP, UDP.
Oct 2025	Flow control mechanisms in TCP; ACKs and timers.
Oct 2025	Connection-oriented versus connectionless; QoS parameters.
Oct 2025	TCP versus UDP in real-time applications; why UDP wins.
Oct 2025	How congestion occurs; how UDP keeps latency low.
Oct 2025 CAE-II	Go Back N, Selective Repeat, piggybacking.

Module 4

Paper	Question
Oct 2026 CAE-II	HTTP, FTP and SMTP in supporting web, file transfer and email.
Oct 2026 CAE-II	Propose a security solution: cryptography, firewalls, VPNs.
Oct 2026 CAE-II	DNS and DHCP: name resolution and dynamic addressing.
Oct 2025	Define cryptography; plaintext, ciphertext and key.
Oct 2025	Application layer function; objective of SSL/TLS.
Oct 2025	HTTP and FTP vulnerabilities and secure alternatives.
Oct 2025	How firewalls filter packets; DNS resolution step by step.
Oct 2025 CAE-II	Protocol choice for web, email and file transfer; FTP active vs passive.
Sept 2024	Short notes: HTTP, SMTP, UDP, ICMP.

Module 5

Paper	Question
Oct 2026 CAE-II	Autonomous Systems and BGP.
Oct 2026 CAE-II	CDN for a far-away e-commerce site.
Oct 2026 CAE-II	Cloud computing: IaaS, PaaS, SaaS.
Oct 2025	Cloud computing: role, scalability, cost.
Oct 2025	Internet versus intranet; Tier-1, 2 and 3 ISPs.
Oct 2025	Define an ISP and its types; redundancy; CDN definition.

The one thing that is almost certain

CRC. It appears in three of the six papers, and two of them use the identical numbers (data 10011101, generator $x^3 + 1$). Learn the division procedure cold; it is 10 easy marks.

The TCP/UDP comparison. Appears four times in different forms: as a table, as "connection-oriented versus connectionless", as "why UDP for real time", and as a short note. One table answers all four.

FDM versus TDM. Appears three times, always worth 3 marks.

The TCP/IP versus OSI comparison. Appears twice, always worth 10.

Where the marks are

Type	Weight	Preparation
Definition / list	3 marks	The five components, the seven layers, the five QoS parameters, the five NIST characteristics.
Comparison table	3–10 marks	OSI/TCP-IP, circuit/packet, FDM/TDM, TCP/UDP, RIP/OSPF, TLS/IPSec, IaaS/PaaS/SaaS.
Calculation	10 marks	CRC is the only routine calculation. Subnetting and Dijkstra may appear.
Diagram	10 marks	Layered models, encapsulation, TCP and IPv4 headers, three-way handshake, congestion sawtooth.
Scenario	10 marks	The October 2026 style. Propose and justify; never just list products.

Five-day plan, 13 to 17 November

Day	Module	Work
Thu 13 Nov	1	Read Module 1 in full. Do both CRC questions by hand, twice, without looking. Draw the OSI and TCP/IP stacks from memory.
Fri 14 Nov	2	Read Module 2. Do the subnetting examples, then the Dijkstra example, then the fragmentation example.
Sat 15 Nov	3 and 4	Read Module 3, then Module 4. Write the TCP/UDP table from memory; then the TLS handshake in six steps.
Sun 16 Nov	5	Read Module 5. Write the CDN delivery process in five steps and the cloud service model table.
Mon 17 Nov	All	Only the last-minute page of each module, plus the formula sheet in Section 7. Do not read anything new.

Exam Focus

Exam rules. For a comparison question, draw the table first; prose loses marks. For the CRC, show every XOR step; the answer alone scores almost nothing. For a scenario question, name the threat, name the control, and say what the control prevents.

Formula and fact sheet

Module 1

Item	Formula or fact
Nyquist capacity	$C = 2B \log_2 L$ bps
Shannon capacity	$C = B \log_2(1 + S/N)$
PCM bit rate	$R = f_s \cdot n$; voice $8000 \times 8 = 64$ kbps
Quantisation SNR	$\approx 6.02n + 1.76$ dB
GBN window	$N = 2^m - 1$
Selective Repeat window	$N = 2^{m-1}$
Sliding window utilisation	$U = N/(1 + 2a)$, $a = T_p/T_t$
Full utilisation	$N \geq 1 + 2a$
ALOHA throughput	$S = Ge^{-2G}$, max 18.4%
Slotted ALOHA	$S = Ge^{-G}$, max 36.8%
CSMA/CD condition	$T_t \geq 2T_p$
CSMA/CD efficiency	$U = T_t/(T_t + 2T_p)$
Full mesh links	$n(n-1)/2$
Hamming check bits	$2^r \geq m + r + 1$

Module 2

Item	Formula or fact
Host bits	$h = 32 - n$
Addresses per subnet	2^h
Usable hosts	$2^h - 2$
Block size	2^h
Aggregation	2^k contiguous prefixes $\Rightarrow$ subtract k
Fragment offset unit	8 bytes
RIP metric	Hop count, max 15, 16 = unreachable
OSPF metric	Cost $= 10^8$/bandwidth
Dijkstra	$O(n^2)$ or $O((n + e) \log n)$
Bellman-Ford	$D(v, X) = \min_w[c(v, w) + D(w, X)]$

Modules 3 to 5

Item	Formula or fact
Well-known ports	20/21 FTP, 22 SSH, 23 Telnet, 25 SMTP, 53 DNS, 67/68 DHCP, 80 HTTP, 110 POP3, 143 IMAP, 161 SNMP, 443 HTTPS
TCP header minimum	20 bytes; options to 60
UDP header	8 bytes
Effective send window	$\min(\text{rwnd}, \text{cwnd}) -$ bytes in flight
Slow start	Double cwnd per RTT
Congestion avoidance	$+1$ MSS per RTT
On timeout	ssthresh = cwnd/2, cwnd = 1 MSS
On 3 duplicate ACKs	Fast retransmit, halve cwnd
IntServ versus DiffServ	Per-flow reservation versus per-class marking
RSA key relation	$ed \equiv 1 \pmod{\phi(n)}$, $\phi = (p - 1)(q - 1)$
Signature	Hash, then encrypt with the *private* key
TLS objectives	Confidentiality, integrity, authentication
IPSec modes	Transport (payload) and tunnel (whole packet)
DHCP	Discover, Offer, Request, Acknowledge

The eight diagrams to be able to draw

1. The seven-layer OSI stack and the four-layer TCP/IP stack side by side, with PDUs.
2. Encapsulation: message, segment, packet, frame, bits.
3. A frame format for CRC: data, appended zeros, remainder, transmitted codeword.
4. The TCP three-way handshake and the four-way teardown.
5. The IPv4 header as a row of labelled boxes.
6. The congestion sawtooth with slow start and congestion avoidance labelled.
7. The Internet hierarchy: access, Tier 3, Tier 2, Tier 1, IXP, content.
8. CDN delivery: DNS picks the nearest edge, cache hit or miss, origin.

Common mistakes that cost marks

1. Using ordinary binary subtraction in a CRC division. It is XOR.
2. Forgetting to append n zeros before dividing in CRC, where n is the generator's degree.
3. Swapping the TCP/IP and OSI layer counts.
4. Saying "FDM divides time". FDM divides frequency; TDM divides time.
5. Describing an IXP as something that routes.
6. Saying Tier-1 ISPs buy transit. They do not.
7. Confusing flow control (rwnd, protects the receiver) with congestion control (cwnd, protects the network).
8. Writing a three-way teardown. TCP teardown is four-way.
9. Signing with the public key. Signing uses the private key.
10. Giving a comparison as two paragraphs instead of a table.

11. Answering a scenario question with a product list and no justification.
12. Forgetting that usable hosts $= 2^h - 2$, not 2^h.

Every calculation in this set was computed and checked programmatically. The CRC remainders, the Dijkstra distances, the subnet boundaries and the RSA values are all verified.